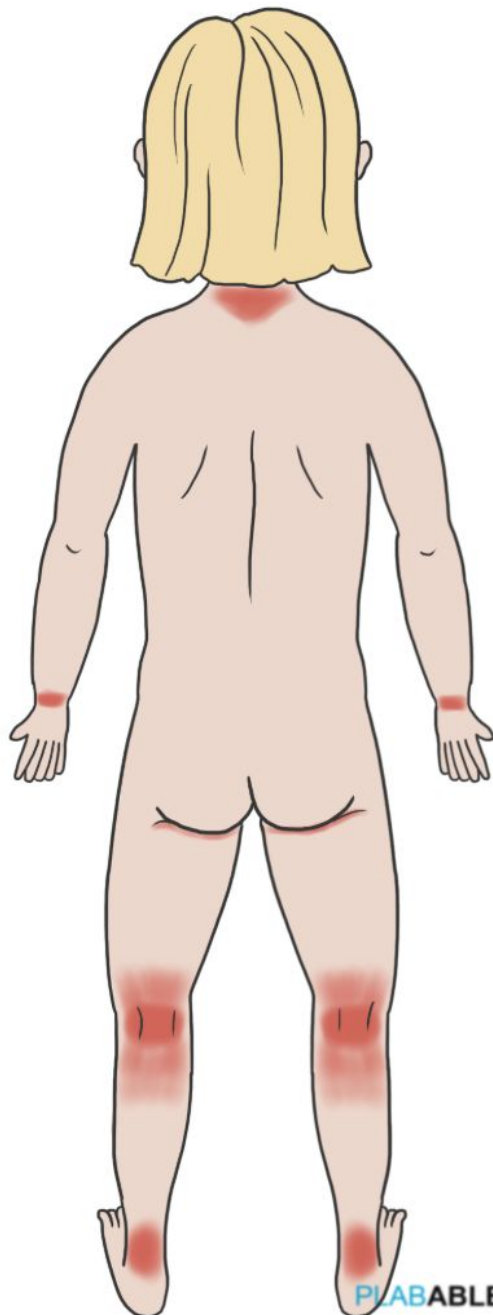


PLABABLE

GEMS

VERSION 1.2

DERMATOLOGY



Cellulitis

Infection of the dermis and subcutaneous tissues with poorly demarcated borders

Presentation

- Erythema
- Warmth
- Swelling and pain of the affected limb
- Most common in the lower limb

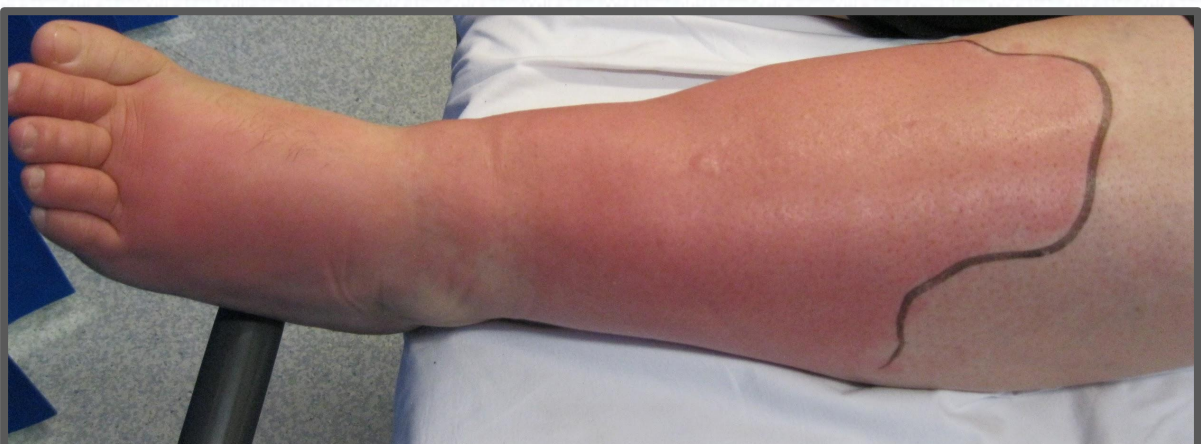
Common organisms: ***Streptococcus*** and ***Staphylococcus***

Risk factors

Diabetes and immunodeficiency

Management

- First line: flucloxacillin 500mg QDS
- Penicillin allergy: clarithromycin 500mg BD
- MRSA +: vancomycin
- Analgesics and limb elevation



Erysipelas

Infection of the dermis and upper subcutaneous tissues with **sharply demarcated borders**

Presentation

- Burning sensation and pruritus of the affected area
- Fiery-red, indurated, tense and shiny plaque
- Common organisms: ***Streptococcus*** and ***Staphylococcus***
- Common areas affected **face** → leg → arm

Management

- **Flucloxacillin** 500mg QDS (first line)
- Erythromycin or clarithromycin (if penicillin allergy)
- Analgesics and limb elevation



Lichen Planus

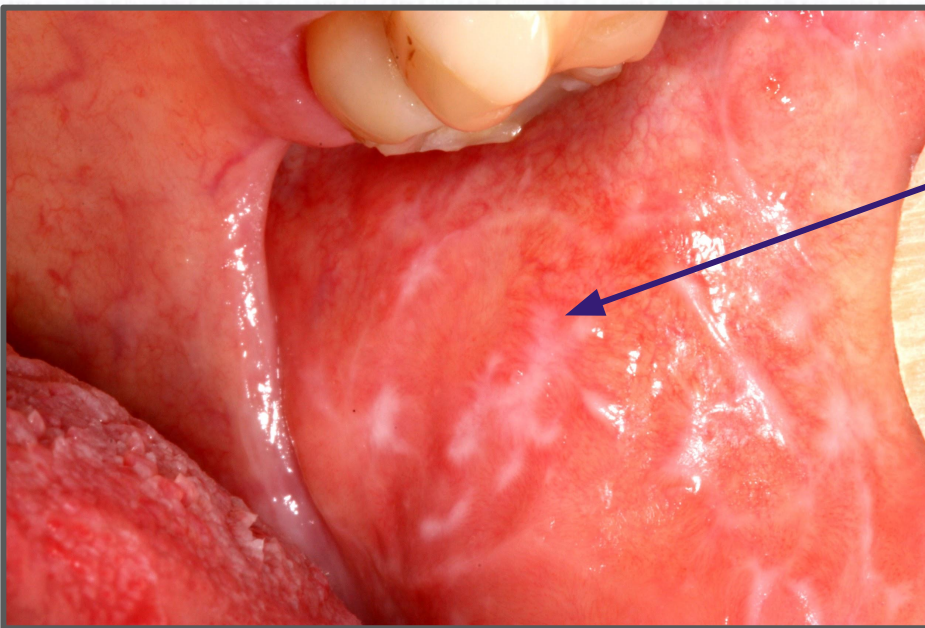
Presentation: 4P

- Pruritic
- Purple
- Papular
- Polygonal rash

Commonly seen on the flexor surfaces and mucosa
(white lacy pattern)

Management (symptomatic)

- Itching - topical steroids and antihistamines



Lacy
White
Pattern



Lesions on
both shins

Urticaria (hives)

Presentation

- Skin rash with red, raised and itchy bumps

Triggers

- Allergies - food, drugs, bee sting
- Skin contact with latex or metals
- Physical stimuli - rubbing



Management

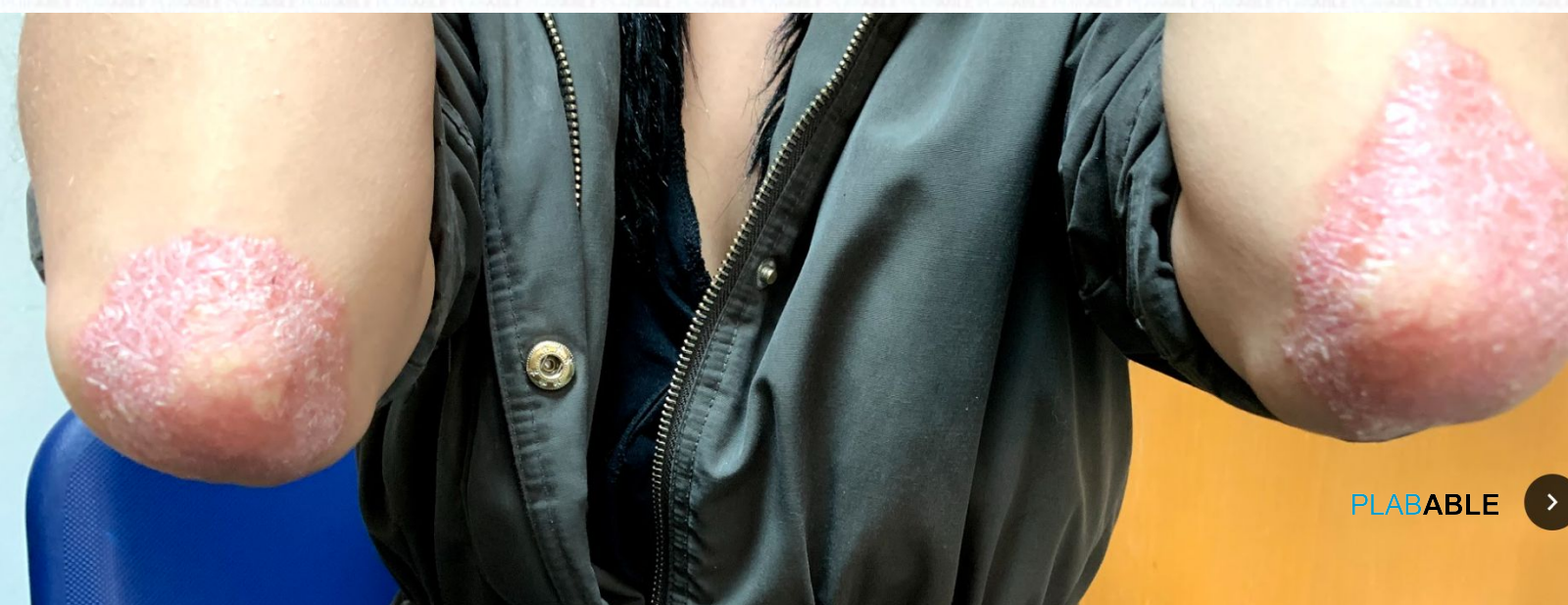
- Antihistamines: cetirizine or loratadine (non-sedating)
- Calamine lotion
- Oral prednisolone (in severe cases)



Psoriasis

Presentation

- Itchy, well-demarcated red/pink elevated plaques
- Associated with overlying white or silvery scales
- **Symmetrical** distribution over extensor surfaces and the scalp
- **Auspitz' sign** - scraping of the lesion causes pinpoint bleed
- Chronic and relapsing



Psoriasis

Management

- **First line**
 - Topical corticosteroids
 - Vitamin D analogues
- **Second line**
 - Phototherapy - **PUVA** (psoralen with UVA)
 - Acitretin
 - Methotrexate

Psoriatic arthritis

- **Symmetric polyarthritis**
(rheumatoid pattern but **RF negative**)
- DIP more commonly involved
- **Treatment:**
 - NSAIDs
 - DMARDs - methotrexate, sulfasalazine and leflunomide

Atopic Dermatitis or Eczema

Presentation

- Itchy red rash affecting skin creases (flexures) such as wrist, elbow folds, behind the knees
- Triggered by allergens such as detergents, dust mites, and pollens
- Associated with hay fever or asthma
- Chronic and relapsing

Management

- Emollients
- Topical steroids (apply ~20 mins after emollient)



Seborrheic Dermatitis

Presentation

- Benign scaling rash seen predominantly in areas with sebaceous glands:
 - Face
 - Scalp (Looks like dandruff)
 - Chest
- Inflammatory reaction to *Malassezia* spp

Management

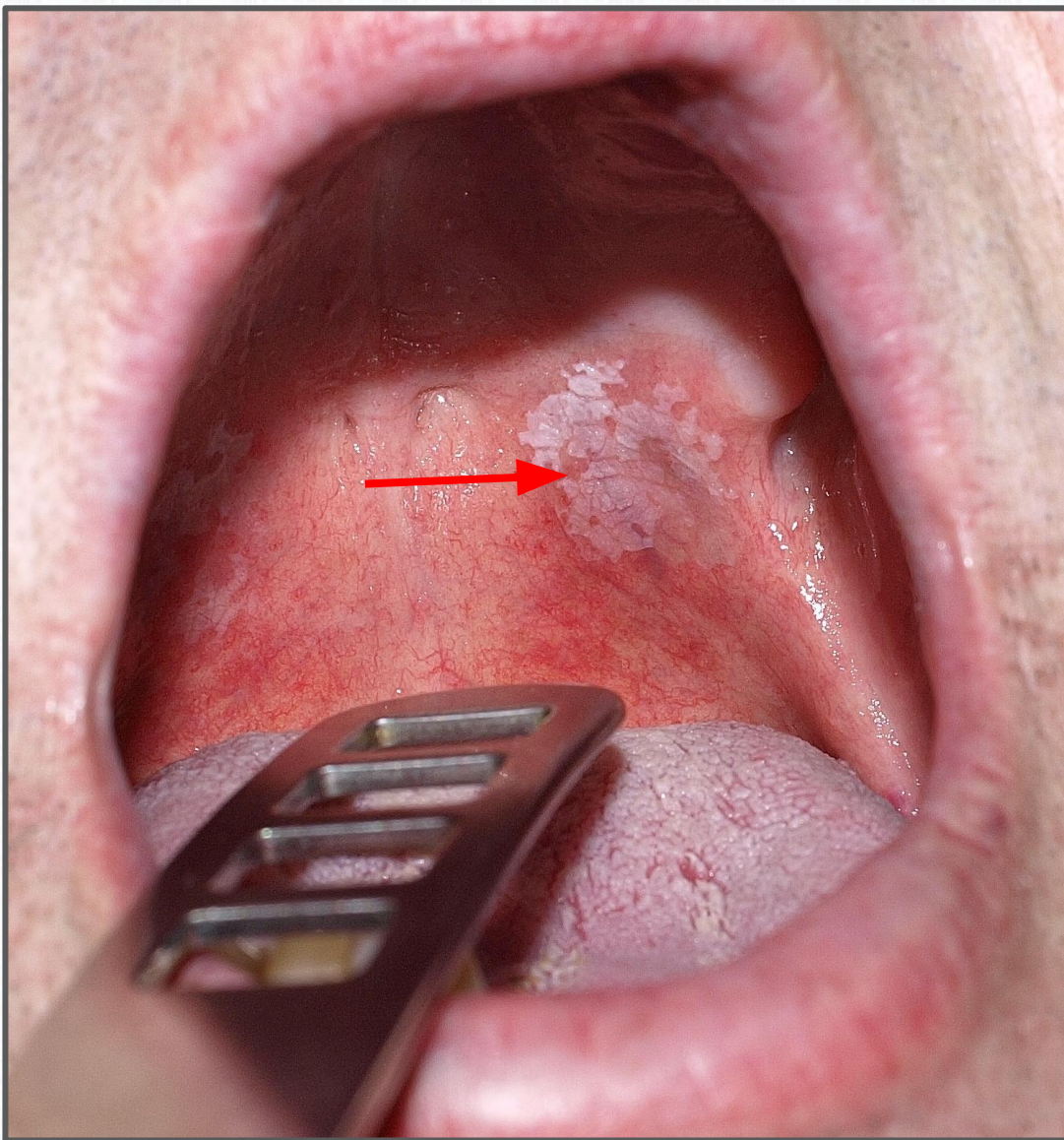
- Medicated shampoo (**ketoconazole** or **selenium sulfide**)
- Steroid cream (in severe cases)



Leukoplakia

Features

- White patch adherent to the oral mucosa and cannot be removed (easily removable in candidiasis)
- Pre-malignant condition due to chronic irritation



Erythroplakia

- Erythematous patch on oral mucosa
- Premalignant lesion

Oral Candidiasis

Curd-like white patches in the tongue which can be wiped off

Risk factors

- Recent antibiotic use
- Immunocompromised state (HIV and chemotherapy)
- Diabetes mellitus
- Cushing's disease
- Inhaled steroids for asthma

Treatment

- Miconazole oral gel (mild)
- Oral fluconazole (severe)



Lichen sclerosus

Presentation:

- Itching worsening at night
- **White thickened patches** seen in the anogenital region in women, and the glans penis and foreskin in men
- Autoimmune chronic inflammatory dermatosis

Treatment:

- Topical clobetasol propionate

Not to be confused with:

Atrophic vaginitis

- **Dyspareunia**
- Vaginal itching or dryness
- Urinary urgency
- Burning pain during micturation
- Seen following **menopause** in elderly women
- **Management:** Topical estrogen

Malignant Melanoma

Cancerous growth of the melanocytes

Presentation

- **A** - Asymmetry
- **B** - Border irregularity
- **C** - Colour irregularity
- **D** - Diameter > 7mm
- **E** - Evolving

Risk factors

- Sun exposure
- White skin > dark skin people

Management

- Surgical excision - early stages
- Radiation and palliative care for late stages

Plabable tip: Think of malignant melanoma if the patient says that the **mole is growing** suddenly and has an occupation which requires **high sun exposure** like a **farmer**

Malignant Melanoma



Which of the following is a bad prognosis following malignant melanoma excision?

- Diameter > 7mm
- Colour variation
- **Depth of invasion**

Malignant Melanoma

Brain trainer:

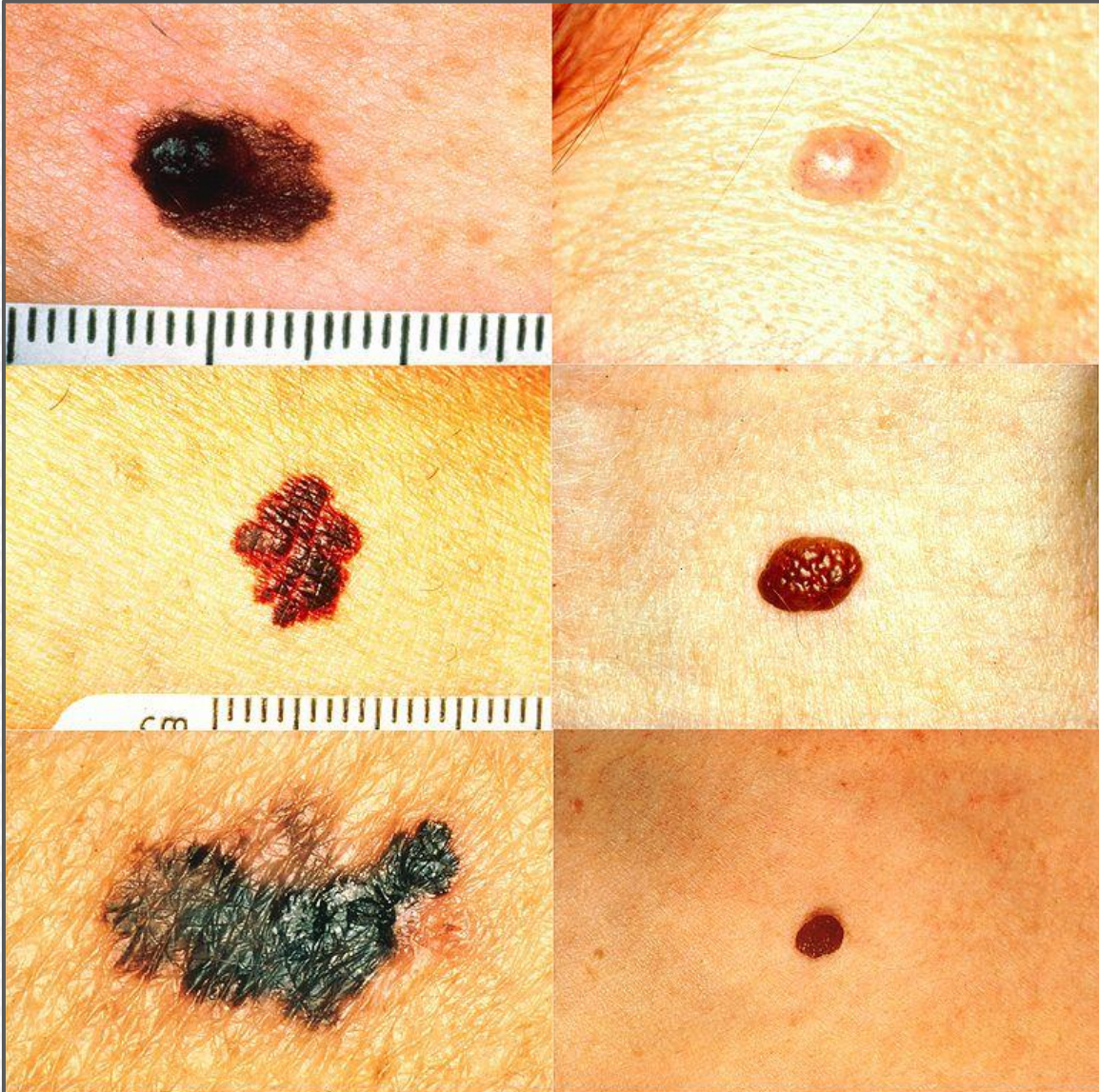
At the time of diagnosis of a malignant melanoma, what parameter is most strongly correlated with prognosis?

➔ **Depth of invasion (Breslow thickness)**

Benign Mole

Malignant
melanoma

Benign mole



Scenario:

A patient comes to the GP clinic with benign mole. It does not interfere with his day to day activities, but he wants it to be removed. What should a GP do?

➔ Referral to private dermatology clinic

Basal Cell Carcinoma

Features

- Most common skin cancer
- Head and neck are most commonly involved
- Initially **Pearly lesion** → **Rodent ulcer** with an indurated edge and ulcerated centre
- Sun exposure is a risk factor

Management

- Surgical excision
- Mohs' Micrographic surgery



Molluscum Contagiosum

Presentation

- Firm, smooth and **umbilicated papules**
- Caused by a pox virus (MCV)
- Immunocompromised people are at high risk

Management

- Reassurance (most resolve in 18 months)
- Cryotherapy



Acne Rosacea

Presentation

- Recurrent episodes of **facial flushing**
- Persistent erythema, telangiectasia, papules and pustules
- **Rhinophyma** (enlarged nose)
- Females > males

Management

- Avoidance of triggers
- Topical metronidazole (first line)
- Oral oxytetracycline or doxycycline for moderate to severe cases



Note: Comedones are absent in Rosacea but present in acne vulgaris

Acne Vulgaris

Features

- Blocked sebaceous gland in face and trunks (**comedones**)
- Seen after **puberty** due to ↑ sebum production and colonisation by ***Propionibacterium acnes***

Management

Mild to moderate:

- Topical **benzoyl peroxide** or **clindamycin**

Moderate to severe:

- Topical **adapalene** (retinoids)
- Oral Doxycycline
- Oral **isotretinoin** (retinoids)

Note:

- Oral isotretinoin is teratogenic
- Women in childbearing age should follow double contraception when on treatment

Acne Vulgaris

Brain trainer:

What medication may be indicated to treat severe or treatment-resistance acne?

→ **Isotretinoin (Vitamin A derivative)**



Acne Vulgaris

Brain trainer:

In the treatment of acne, benzoyl peroxide is intended to act primarily against which species of bacteria?

→Propionibacterium

Fungal Infections

Tinea Capitis

- Scalp ringworm is caused by the dermatophytes *Microsporum* spp and *Trichophyton* spp
- Common cause: **alopecia**
- Treatment: Oral griseofulvin and terbinafine



Tinea Cruris:

- Fungal infection of the groin
- Lesion is erythematous with central clearing and raised edge
- Treatment: Topical clotrimazole or terbinafine



Impetigo

Features

- Superficial bacterial skin infection causing **honey-coloured crusted plaques**
- Common organisms: *S. aureus* & *S. pyogenes*

Risk factors

- Atopic eczema
- Insect bites
- Trauma to skin

Common in children

Treatment

- Topical **fusidic acid** or mupirocin (if localised)
- Oral flucloxacillin (in extensive cases)



Complications

Group A **beta-haemolytic streptococci** can cause:

- Scarlet fever
- Glomerulonephritis

Erythema Multiforme

Presentation

- Hypersensitivity reaction caused by **infection** or **drugs**
- Erythema (redness), multiforme (many shapes)
- Reaction ranges from minor to major
- **Infections** (common cause):
 - HSV 1 and 2
 - *Mycoplasma pneumoniae*
- **Drugs:** penicillins, sulfonamides, anticonvulsants

Management

- Withdrawal of the drug
- Treatment of infection if any

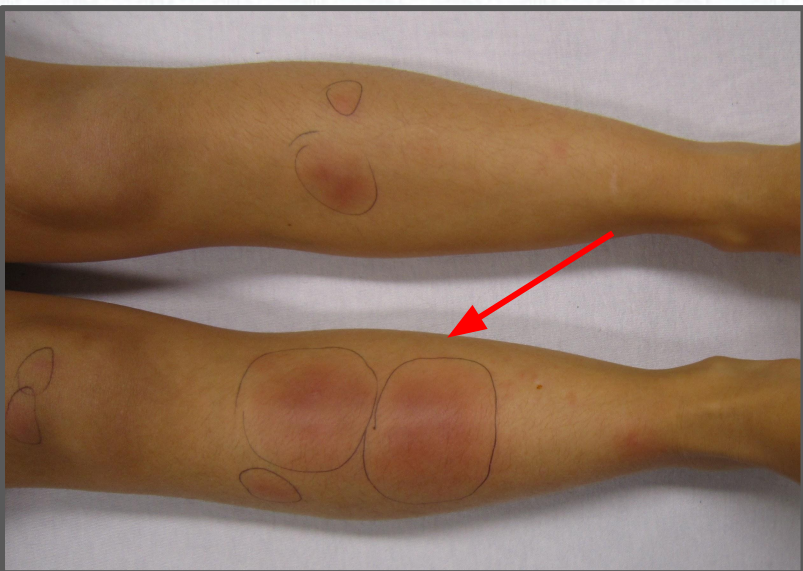


Target
lesion

Erythemas

Erythema Marginatum	Rheumatic fever
Erythema Nodosum	Painful tender nodules over shins Causes: IBD, penicillins, sarcoidosis, TB
Erythema Infectiosum (Fifth disease)	Parvovirus B19 Slapped cheek appearance
Erythema Migrans	Bull's eye lesion seen in Lyme disease - <i>Borrelia burgdorferi</i> Treat: doxycycline or amoxicillin (pregnant)

Erythema Nodosum



Erythema Migrans



Dermatitis Herpetiformis

Presentation

- **Very itchy** burning blisters which are often found on extensor surfaces

Management

- Short term: dapsone
- Long term: life-long gluten-free diet



Erythema ab igne

Etiology

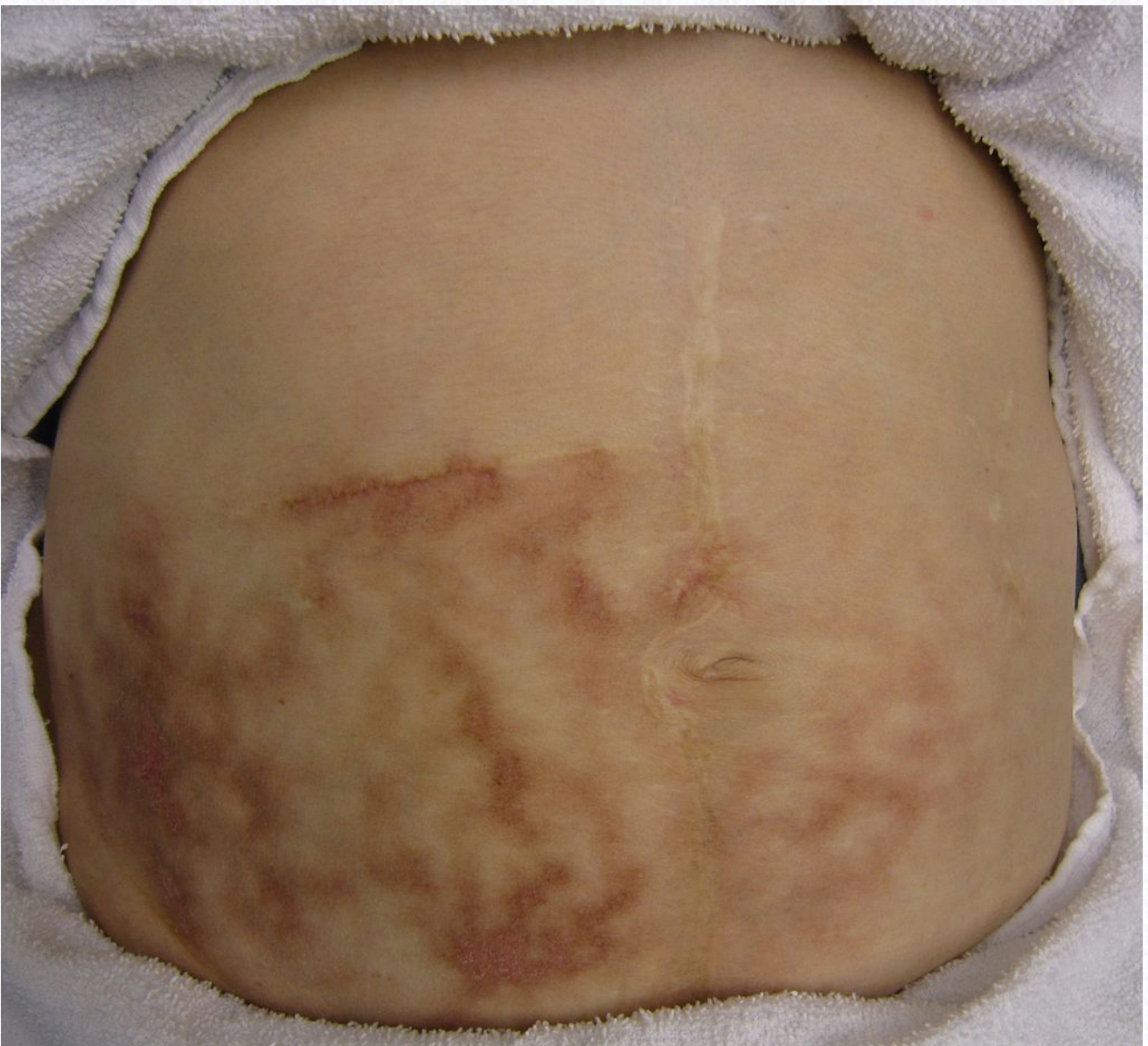
- Chronic exposure to thermal heat which is insufficient to cause a burn.
- Also known as “hot water bottle rash”

Presentation

- Redness, hyperpigmentation, ± itching/burning

Management

- Avoidance of thermal heat



Polymorphic Eruption of Pregnancy

Presentation

- Itchy urticaria-like rash, raised lumps and inflamed skin on abdomen. Classic feature is sparing of the umbilicus. Usually occurs in first pregnancy and third trimester.

Management

- Emollients
- Steroids



Mongolian Blue Spot - Dermal Melanosis

Presentation

- **Bluish discoloration** over the base of the back and buttocks
- Benign and congenital

Management

- Reassurance as this usually fade after few years



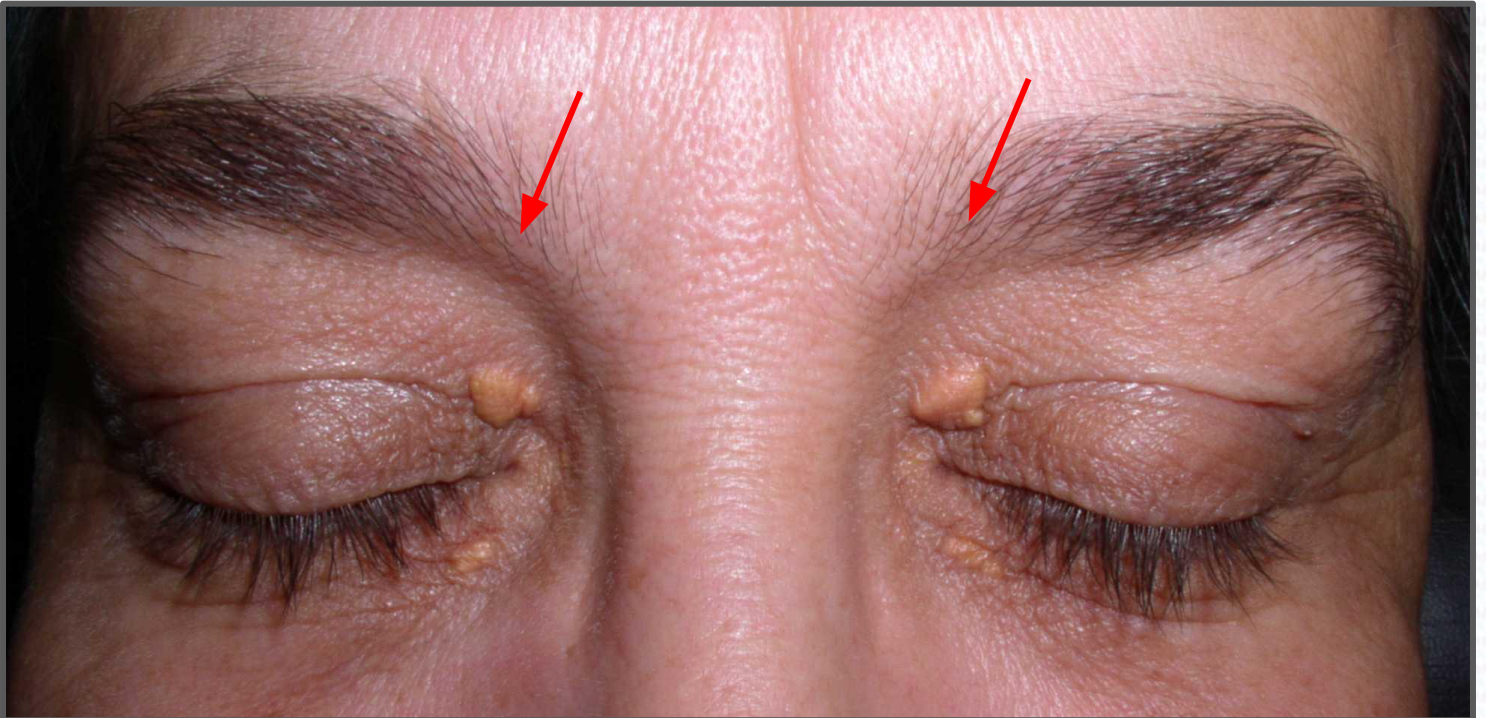
Xanthelasma

Presentation

- **Yellow flat plaques** over the upper or lower eyelids (common near the inner canthus)
- Associated with hyperlipidemia

Management

- Surgical excision only for cosmetic reasons
- Lipid profile to analyse cardiac risk



Note:

- **Xanthochromia** - yellowish discoloration of the CSF
- Seen in **subarachnoid haemorrhage** as heme is converted into bilirubin

Herpes Zoster

Feature

- HHV infection in childhood → virus lies dormant in sensory nervous system → reactivation → eruptive skin lesions and pain in one particular dermatome

Treatment

- Oral **acyclovir** or **valacyclovir**
- NSAIDs for pain



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